Doc. 264 – Ferrotoroidicity and Quaternary Memory: All Formulas Reinterpreted from the FFGFT Perspective

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Preamble: Why This Paper Matters for FFGFT

Qureshi et al. (2026, *Nature Communications* 17:4033) report the first experimental demonstration of quaternary (4-state) non-volatile memory in a purely antiferromagnetic single crystal of $LiNi_{0.8}Fe_{0.2}PO_4$, using *ferrotoroidic* domain control via perpendicular electric and magnetic fields. The four distinct magnetic domains arise from a spontaneous spin rotation below a phase transition temperature $T_4 = 21$ K, and their selective population is verified by Spherical Neutron Polarimetry (SNP).

From the FFGFT perspective, this paper is remarkable for three independent reasons:

1. The *torus* is not only the fundamental geometric object of FFGFT (T^4, Dok. 001/002) but appears directly as the physical origin of the 4-state structure in the experiment: four domain states = two toroidal directions $\times$ two orientation states, a 2×2 decomposition that mirrors the mode labelling on a compact torus.
2. The *polarization matrix formalism* (Blume-Maleev equations) is a rotation-matrix representation of state evolution – structurally identical to the unitary operators on the T^4 mode space in FFGFT (Dok. 203, Recursion $\mathscr{R}$).
3. The *ferrotoroidic order parameter* breaks both space-inversion and time-inversion symmetry simultaneously – exactly the symmetry structure of the FFGFT winding modes, which carry both a spatial winding number and a temporal orientation.

Status of this document. Every formula from the paper is reproduced and reinterpreted. The FFGFT interpretations are reduction layer unless explicitly marked as bridge or core. The paper's own results are not contested; the reinterpretation adds a geometric reading, not a physical correction.

.1 Toroidization Formula (Eq. 1 in the Paper)

The formula

$$\mathbf{T} = 2S\frac{\varepsilon_a}{\Omega}\cos\varphi \cdot \mathbf{b}\times\mathbf{c} \qquad \text{(Paper Eq. 1)}$$

where S is the spin magnitude, ε_a/Ω is a geometric prefactor (unit cell volume Ω, spin displacement ε_a), φ is the rotation angle of the magnetic moments in the a–b plane, and $\mathbf{b}\times\mathbf{c}$ defines the toroidal direction along the crystallographic c axis.

FFGFT interpretation

In FFGFT, the torus T^4 has four compact dimensions. The toroidization **T** is the macroscopic projection of a *winding flux*:

- S corresponds to the winding quantum number n of a torus mode (Dok. 001, fundamental relation $T\tilde{m} = 1$): the spin magnitude measures how many times the mode winds around the compact direction.
- ε_a/Ω is the *packing density* of the winding in the unit cell – directly analogous to the packing deficit $\xi = 4/30000$ in FFGFT, which measures how much the 3D packing of space deviates from a smooth manifold (Dok. 009).
- $\cos\varphi$ is the *projection* of the winding onto the observable direction. In FFGFT, this corresponds to the decompactification projection: the full T^4 winding is unit-free (Layer 1), and $\cos\varphi$ represents the fraction that becomes observable in the SI-projected (Layer 2) description.
- $\mathbf{b}\times\mathbf{c}$ defines the toroidal axis – in FFGFT, this is the projection direction from the 4D compact space onto the 3D observable subspace. The cross product structure is the same as in the T^4 volume form restricted to a 2-cycle.

Key observation: **T** is independent of the rotation angle φ in the sense that its *direction* (sign) does not change as φ varies – only its magnitude changes. This is structurally identical to the FFGFT non-closure theorem (Dok. 193): the topological charge (winding number) is invariant under continuous deformations of the mode within a winding class.

.2 Polarization Matrix (Eq. 2 and Eq. 3)

The formula

$$\mathbf{P}_f = \mathbf{P}\,\mathbf{P}_i \qquad \text{(Paper Eq. 2)}$$

$$\mathscr{P} = \frac{1}{I}\begin{pmatrix} N^2 - M_\perp^2 & J_{nz} & J_{ny} \\ J_{nz} & N^2 + M_{\perp y}^2 - M_{\perp z}^2 & R_{yz} \\ J_{ny} & R_{yz} & N^2 - M_{\perp y}^2 + M_{\perp z}^2 \end{pmatrix} \qquad \text{(Paper Eq. 3)}$$

where N is the nuclear structure factor, $M_\perp$ is the magnetic interaction vector (perpendicular component of the magnetic structure factor), $J_{ni} = 2\,\mathrm{Im}(NM_{\perp i}^*)$ are nuclear-magnetic interference terms, and $R_{yz} = 2\,\mathrm{Re}(M_{\perp y}M_{\perp z}^*)$.

FFGFT interpretation

The transformation $\mathbf{P}_f = \mathbf{P}\,\mathbf{P}_i$ is a *rotation matrix acting on the neutron spin state*. In FFGFT, this is the direct analogue of the recursion operator $\mathscr{R}$ (Dok. 203) acting on the T^4 mode vector: the state vector transforms under a unitary rotation determined by the geometric structure of the scattering object.

The matrix $\mathscr{P}$ has four distinct forms for the four domains – each domain corresponds to a *different winding configuration* on the torus, and each winding configuration produces a different rotation of the mode vector. This is structurally identical to the four winding classes in FFGFT: two toroidal directions (time-reversal pairs, $\pm$ winding) $\times$ two orientation states (spatial reflection pairs), giving a $\mathbb{Z}_2 \times \mathbb{Z}_2$ group structure.

The nuclear-magnetic interference terms J_{ni} are non-zero because the magnetic structure factor $M_\perp$ is *purely imaginary* in the antiferromagnetic phase (the moments related by inversion are antiparallel). In FFGFT, imaginary structure factors correspond to modes that are *anti-phase* on opposite sides of the torus – precisely the antiferromagnetic (head-to-tail) configuration that defines the ferrotoroidic ground state.

The diagonal element $N^2 - M_\perp^2$ measures the imbalance between nuclear (scalar) and magnetic (vector) scattering – in FFGFT, this is the imbalance between the scalar field E (Lagrangian $\mathscr{L} = \xi(\partial E)^2$, Dok. 010) and the vector field components, which determines the energy density $T_{00} = \xi[(\partial_t E)^2 + (\nabla E)^2]$.

.3 General Polarization Matrix Including Beam Imperfections (Eq. 9)

The formula

$$\mathscr{P} = \begin{pmatrix} \frac{p_{fx}}{p_{ix}}\frac{N^2-M_\perp^2-J_{yz}}{I_x} & \frac{p_{fx}}{p_{iy}}\frac{J_{nz}-J_{yz}}{I_y} & \frac{p_{fx}}{p_{iz}}\frac{J_{ny}-J_{yz}}{I_z} \\ \frac{p_{fy}}{p_{ix}}\frac{J_{nz}+R_{ny}}{I_x} & \frac{p_{fy}}{p_{iy}}\frac{N^2+M_{\perp y}^2-M_{\perp z}^2+R_{ny}}{I_y} & \frac{p_{fy}}{p_{iz}}\frac{R_{yz}+R_{ny}}{I_z} \\ \frac{p_{fz}}{p_{ix}}\frac{J_{ny}+R_{nz}}{I_x} & \frac{p_{fz}}{p_{iy}}\frac{R_{yz}+R_{nz}}{I_y} & \frac{p_{fz}}{p_{iz}}\frac{N^2-M_{\perp y}^2+M_{\perp z}^2+R_{nz}}{I_z} \end{pmatrix} \qquad \text{(Paper Eq. 9)}$$

where p_{fx}, p_{fy}, p_{fz} are the final analysis efficiencies and p_{ix}, p_{iy}, p_{iz} are the initial polarization values along the three reference axes; $R_{ni} = 2\,\mathrm{Re}(NM_{\perp i}^*)$, $J_{yz} = 2\,\mathrm{Im}(M_{\perp y}M_{\perp z}^*)$, and the scattered intensities I_x, I_y, I_z now include polarization-dependent corrections.

FFGFT interpretation

The factors p_{fx}/p_{ix} etc. are *projection efficiencies* – they measure how accurately the measurement apparatus resolves the polarization state. In FFGFT, these correspond to the *decompactification factors*: the ratio between the observable (SI-projected, Layer 2) quantity and the underlying geometric (unit-free, Layer 1) quantity. Perfect measurement ($p_{fi} = 1$) corresponds to the Layer 1 limit (no embedding price); real measurement ($p_{fi} < 1$) corresponds to the Layer 2 projection with its inherent incompleteness.

The quantities $R_{ni} = 2\,\mathrm{Re}(NM_{\perp i}^*)$ are *real* nuclear-magnetic interference terms, while J_{ni} are imaginary ones. In FFGFT, this separation is exactly the decomposition into *even* (cosine, time-symmetric) and *odd* (sine, time-asymmetric) winding mode components (Dok. 203, Recursion $\mathscr{R}$). The real part R_{ni} involves modes that are symmetric under time reversal; the imaginary part J_{ni} involves modes that change sign under time reversal – exactly the ferrotoroidic modes that break time-inversion symmetry.

The general intensity $I_i = N^2 + M_\perp^2 + p_{ii}J_{yz}$ (or similar with R_{ni}) shows that the observable intensity depends on the *initial polarization* – in FFGFT, this corresponds to the fact that the observable energy density T_{00} depends on the temporal component $(\partial_t E)^2$: a field with no temporal variation has $T_{00} = 0$ (Dok. 010), just as a completely unpolarized neutron beam ($p_{ii} = 0$) gives zero interference terms.

.4 Dzyaloshinskii–Moriya Hamiltonian (Eqs. 4–6)

The formulas

$$\mathcal{H}_1^{\mathrm{DM}} = \mathbf{D}_{12} \cdot (\mathbf{S}_1 \times \mathbf{S}_2) + \mathbf{D}_{34} \cdot (\mathbf{S}_3 \times \mathbf{S}_4) \quad \text{(Paper Eq. 4)}$$

$$\mathcal{H}_{1c}^{\mathrm{DM}} = D_{12}^c (S_1^a S_2^b - S_1^b S_2^a - S_3^a S_4^b + S_3^b S_4^a) \quad \text{(Paper Eq. 5)}$$

$$\mathcal{H}_{1b}^{\mathrm{DM}} = -D_{12}^b (S_1^a S_2^c - S_1^c S_2^a - S_3^a S_4^c + S_3^c S_4^a) \quad \text{(Paper Eq. 6)}$$

with $\mathbf{D}_{34} = -\mathbf{D}_{12}$ (antisymmetry under inversion), and the components D_{12}^b, D_{12}^c being the projections of the Dzyaloshinskii–Moriya (DM) vector onto the crystallographic b and c axes.

FFGFT interpretation

The Dzyaloshinskii–Moriya interaction is an *antisymmetric exchange*: $\mathbf{D} \cdot (\mathbf{S}_1 \times \mathbf{S}_2)$ is odd under exchange of spins 1 and 2 (compare: the symmetric Heisenberg exchange $J\mathbf{S}_1 \cdot \mathbf{S}_2$ is even). In FFGFT, antisymmetric interactions correspond to the *non-commutative* products of winding modes – the cross-product structure $S_i \times S_j$ is precisely the structure that the Non-Closure Theorem (Dok. 193) identifies as the topologically constrained product of mode labels from different winding classes.

The relation $\mathbf{D}_{34} = -\mathbf{D}_{12}$ (due to inversion symmetry) corresponds to the FFGFT symmetry that modes related by spatial inversion have opposite winding numbers: if mode n winds clockwise, the inversion- related mode $-n$ winds anti-clockwise. This is the T^4 analogue of the crystallographic inversion operator.

The components D^c and D^b (with $|D^b| < |D^c|$ from the geometry, as shown in Fig. 6 of the paper) reflect that the DM vector is not aligned with any crystallographic axis – it lies at 35° from the c axis in the b–c plane. In FFGFT, this corresponds to the *anisotropic* ξ*-coupling*: the geometric parameter ξ enters differently in different compact directions, with the dominant coupling in the direction of the largest winding mode (here, the c-axis ferrotoroidic direction).

The explicit form of $\mathcal{H}_{1c}^{\mathrm{DM}}$ (Eq. 5) is a sum of four terms $\pm S_i^a S_j^b$ – the ab-plane spin cross product. In FFGFT language, this is the *mixed-mode bilinear* that couples the a-axis winding (orientation domain) to the b-axis winding (primary ordered moment): exactly the coupling between two different T^4 compact directions that produces the 2×2 domain structure.

.5 Polarization Tensor Element Formula (Eq. 7)

The formula

$$P_{fi} = \frac{n_{fi}^{++} - n_{fi}^{+-}}{n_{fi}^{++} + n_{fi}^{+-}} \quad \text{(Paper Eq. 7)}$$

where n_{fi}^{++} (n_{fi}^{+-}) counts neutrons with initial spin parallel to i and final spin parallel (antiparallel) to f.

FFGFT interpretation

This is the standard quantum-mechanical expectation value of the polarization operator, normalized by the total count. In FFGFT, it corresponds to the ratio of two mode populations:

modes in the "+" winding state vs. modes in the "−" winding state, after scattering (i.e., after the recursion operator $\mathscr{R}$ has acted once).

The normalization by $n^{++} + n^{+-}$ (total count) is the FFGFT *decompactification*: the absolute number of modes (a Layer 1 quantity) is not directly observable; only the ratio (a dimensionless, Layer 1-compatible quantity) is. This is the same structure as the FFGFT T_{CMB} relation: only the ratio $T_{\text{CMB}}/E_\xi = (16/9)\xi^2$ is a pure geometric prediction; the absolute Kelvin value requires the SI embedding.

.6 Final Polarization Including Created/Annihilated Components (Eq. 8)

The formula

$$\mathbf{P}_f = \mathscr{P}'\,\mathbf{P}_i + \mathscr{P}'' \qquad \text{(Paper Eq. 8)}$$

where $\mathscr{P}'$ is the purely rotational part of the polarization transformation and $\mathscr{P}''$ is the created or annihilated polarization (non-zero when the scattering process itself creates polarization).

FFGFT interpretation

The decomposition into a *rotational* part $\mathscr{P}'$ and a *creation* part $\mathscr{P}''$ maps directly onto the FFGFT distinction between two types of mode transformation:

- $\mathscr{P}'\,\mathbf{P}_i$: *coherent* transformation – the input state is rotated without energy exchange. This corresponds to the FFGFT recursion $\mathscr{R}$ (Dok. 203) acting on an existing mode: the winding number is preserved, only the phase (orientation) changes.
- $\mathscr{P}''$: *spontaneous* polarization – new polarization is created by the scattering process itself. This corresponds to *mode creation* in FFGFT: a new mode is generated at a T^4 node (Dok. 204, Fractal Boundary Condition), with a winding number determined by the local geometry rather than the input state.

In the specific magnetic structure of $LiNi_{0.8}Fe_{0.2}PO_4$, $\mathscr{P}'' = 0$ because the magnetic structure factors are purely imaginary (as noted in the Methods section of the paper). In FFGFT, this corresponds to the case where the T^4 geometry is *static*: no new modes are created spontaneously, and all transformations are purely rotational (unitary). This is the T^4 static ground state – precisely the FFGFT cosmological picture (static universe, no spontaneous mode creation = no expansion).

.7 The Four-Domain Structure: 2×2 on T^4

The paper's central experimental result is that four distinct magnetic domains exist, controlled by two independent handles:

1. **Toroidal domains** (2 states): selected by $\mathbf{E} \times \mathbf{H}$ (parallel or antiparallel to c).
2. **Orientation domains** (2 states): selected by H_c (a small c-component of the magnetic field, via the DM energy).

The four combined states $(a_4', a_4'', b_4', b_4'')$ are related by:

- Time-inversion $1'$: maps $a_4 \leftrightarrow b_4$ (toroidal flip).
- 2-fold screw axis $2_1 \| y$: maps $' \leftrightarrow ''$ (orientation flip).

FFGFT interpretation

This $\mathbb{Z}_2 \times \mathbb{Z}_2$ group structure is the direct experimental realization of the T^4 winding mode classification in FFGFT:

- The *toroidal* $\mathbb{Z}_2$: corresponds to the two possible orientations of a closed winding loop on the T^4 (clockwise vs. anti-clockwise = $\pm$**T**). This is the fundamental "arrow of time" degree of freedom in FFGFT: the intrinsic time $\tilde{T}$ can point in two directions on the compact torus (Dok. 230, Bell correlations as T^4 geometry).
- The *orientation* $\mathbb{Z}_2$: corresponds to the two possible spatial orientations of the winding plane within the 4D torus. In FFGFT, this is the spatial parity of the mode – whether the winding wraps in the $+c$ or $-c$ spatial direction.

The four combined states are thus:

$$\{+\tilde{T}, +c\}, \quad \{+\tilde{T}, -c\}, \quad \{-\tilde{T}, +c\}, \quad \{-\tilde{T}, -c\}$$

which is exactly the FFGFT mode labelling for a T^4 with one temporal and one spatial compact direction contributing to the observable domain structure.

WBE connection: The four-domain structure has an effective dimensionality $D_{\text{eff}} = 2$ (two binary degrees of freedom). The WBE exponent for this effective dimension is $\alpha = D_{\text{eff}}/(D_{\text{eff}} + 1) = 2/3$. This is neither $w = -3/4$ (Onur's 3D network) nor $w = -1$ (ΛCDM), but an intermediate value corresponding to the 2D nature of the toroidal domain control surface. This suggests that different effective dimensions apply at different scales and for different physical subsystems – which is consistent with the FFGFT picture that ξ encodes a fractional departure from integer dimension at all scales (Dok. 242, "Die Skalenleiter").

.8 Quaternary Memory and FFGFT Information Structure

The paper shows that 8 quaternary units yield $4^8 = 65536$ combinations, compared to $2^8 = 256$ for binary – a factor of 256 increase. This is the *power-law increase of storage density* enabled by going from binary to quaternary logic.

In FFGFT, this corresponds to the fundamental observation that the T^4 torus carries *more information per mode* than a point or a circle: each mode is labelled by four winding numbers (one per compact dimension), not just one (binary). The storage capacity grows as 4^N (for N modes, each with 4 states) vs. 2^N (binary), which is exactly the ratio between the T^4 phase space and a product of circles.

The *non-volatile* nature of the memory – the domain populations are stable in zero applied field, proven by measuring at $E = H = 0$ – corresponds in FFGFT to the *topological stability* of winding modes: a mode with non-zero winding number cannot decay to zero winding without crossing an energy barrier (the minimal energy of the T^4 boundary condition, Dok. 204). This is why toroidal memory is non-volatile: the topological charge is conserved unless a conjugate field large enough to overcome the anisotropy barrier is applied.

.9 Relation to the FFGFT Parameter ξ

The geometry of the paper involves three key length scales:

- The unit cell dimension (~ 10 Å): sets the scale of the individual winding loop.
- The domain size (~mm): the macroscopic coherence length of the toroidal order.
- The phase transition temperatures $T_4 = 21$ K and $T_2 = 25$ K: the energy scales for the two levels of domain structure.

In FFGFT, the ratio of the smallest and largest scales is governed by ξ:

$$\frac{L_0}{L_{\text{domain}}} \sim \xi^n \quad \text{for some integer or fractional } n.$$

The exact value of n is not determined within this document (reduction layer), but the existence of a ξ-controlled hierarchy between the atomic scale (unit cell) and the macroscopic scale (domain) is a structural prediction of FFGFT.
Energy ratio: The ratio of the two phase transition temperatures is $T_4/T_2 = 21/25 = 0.840$. In FFGFT, ratios of energy scales are related to powers of ξ:

$$1 - \xi = 1 - \frac{4}{30000} \approx 0.99987 \quad \text{(first order)}$$

The observed ratio 0.840 is closer to $1 - \xi^{1/4} = 1 - 0.107 \approx 0.893$ (a rough estimate). A precise geometric derivation of this ratio from the FFGFT T^4 geometry is an open question (reduction layer).

.10 Summary: What the Experiment Reveals about the T^4 Structure

- **Toroidal order is real and controllable:** The experiment provides the first unambiguous microscopic proof of four-state toroidal domain control in a bulk antiferromagnet. This confirms that the T^4 topology – the basis of FFGFT – is not merely a mathematical construct but manifests as a physical domain structure in real materials.
- **The $\mathbb{Z}_2 \times \mathbb{Z}_2$ structure is exact:** The four domains are precisely related by time-inversion and a spatial screw operation – the same group that labels T^4 winding modes in FFGFT.
- **Purely imaginary structure factors = static T^4:** The fact that $M_\perp$ is purely imaginary (antiferromagnetic, head-to-tail configuration) means $\mathscr{P}'' = 0$: no spontaneous polarization is created. This is the FFGFT static ground state.
- **Non-volatility = topological stability:** The memory is non-volatile because the winding number is topologically protected. In FFGFT, this is the same mechanism that stabilizes the lepton mass hierarchy: the winding numbers $p_e = 3/2$, $p_\mu = 2$, $p_\tau = 5/2$ are topologically protected (Dok. 190, P2).

Status of the interpretations in this document. Sections 1–5 (individual formulas) are *reduction layer*: structural analogies between the paper's formalism and FFGFT, without a new quantitative prediction. Sections 6 (4-domain = T^4 modes) and 7 (non-volatility = topological stability) are *bridges*: the correspondence is algebraically exact ($\mathbb{Z}_2 \times \mathbb{Z}_2$ group structure, topological charge conservation), derived from both frameworks independently. Section 8 (ξ-controlled scale hierarchy) is *reduction layer*: the structure is plausible but the exact power of ξ is not determined.

.11 Fractal Path Lengthening vs. Tired Light – a Necessary Distinction

The FFGFT explanation of redshift as *fractal path lengthening* in a static universe (Doc. 190, R14) must be distinguished from the historical *Tired Light* hypothesis (Zwicky 1929), which is rejected in mainstream astrophysics for three reasons. FFGFT avoids all three:

1. **No blurring of distant sources.** Tired Light requires photon scattering off intervening matter, which would broaden wavefronts and blur distant galaxies. Fractal path lengthening involves no scattering: the photon follows the topological geometry of the static

vacuum (the "head-to-tail" winding structure demonstrated in $LiNi_{0.8}Fe_{0.2}PO_4$), preserving full wavefront coherence.

2. **Consistent geometric time dilation** $(1+z)$**.** Tired Light predicts no time dilation of supernova light curves, which is observationally ruled out. In FFGFT, every signal traversing a fractally lengthened path carries the same wavelength ratio λ_b/λ_e in wavelength as in pulse duration – both scale geometrically through the fractal path lengthening, not through energy loss. The ratio is preserved, and the observed supernova time dilation follows as a direct consequence. (The symbol $z = \lambda_b/\lambda_e - 1$ is a label for the measured ratio, not a primary FFGFT quantity, Doc. 190, R14.)
3. **Preservation of the CMB blackbody spectrum.** Photon scattering would distort the spectral shape. Fractal path lengthening scales all frequencies uniformly ($\Delta\nu/\nu =$ const), preserving the perfect blackbody form of the CMB spectrum.

The ferrotoroidic vacuum structure demonstrated in the paper provides the physical mechanism for (1): the "head-to-tail" antiferromagnetic configuration (purely imaginary $M_\perp$, hence $\mathscr{P}'' = 0$, no spontaneous polarization) ensures coherent, scattering-free propagation.

.12 Topological Structure Across Three Scales

Table 1: The same $\mathbb{Z}_2 \times \mathbb{Z}_2$ topological structure appears at three distinct scales.

Scale	System	Topology	States
Atomic (10^{-10} m)	$LiNi_{0.8}Fe_{0.2}PO_4$	Toroidal spin rings	4 domains (SNP)
Sub-Planck (10^{-38} m)	FFGFT vacuum (T^4)	Toroidal winding modes	4 winding classes
Cosmological (Mpc)	Observable universe	Fractal path lengthening	$H_0 = 66.2$ km/s/Mpc (no Λ)

The cosmological row uses H_0 from ξ (Doc. 263/026) as the geometric identification, not an equation-of-state parameter w: FFGFT has no dark energy field and no expansion, so there is no w to assign. The scale invariance of the topological structure is structural, not numerical.

.13 Correspondence Table: Paper ↔ FFGFT

Reference. N. Qureshi et al., *Toroidicity as a route towards non-volatile quaternary memory in antiferromagnets*, *Nature Communications* **17**, 4033 (2026). DOI: 10.1038/s41467-026-70767-8.

Table 2: Verified correspondences between Qureshi et al. (2026) and FFGFT. Only entries with an explicit derivation or structural argument in both frameworks are listed. Speculative or numerically unsupported identifications are excluded.

Nature paper (Qureshi et al. 2026)	FFGFT (T^4 geometry)
Head-to-tail spin rings	Closed winding loops on T^4
Toroidal moment $\mathbf{T} \parallel \hat{\mathbf{c}}$	Winding flux in compact c-direction
Spontaneous spin tilt (φ)	Decompactification projection ($\cos\varphi$, Layer 2)
4 discrete domains	$\mathbb{Z}_2 \times \mathbb{Z}_2$ winding mode classes
Time inversion $1'$: $a \leftrightarrow b$	Temporal winding direction ($\pm\tilde{T}$)
Screw axis $2_1 \| y$: $' \leftrightarrow ''$	Spatial winding parity ($\pm c$)
Polarization matrix $\mathscr{P}$ (rotation)	Recursion $\mathscr{R}$ on T^4 mode vector (Doc. 203)
$\mathscr{P}'' = 0$ (purely imaginary $M_\perp$)	Static T^4 ground state (no mode creation)
Non-volatile memory	Topological charge conservation (Doc. 204)
Measurement ratio P_{fi} (dimensionless)	Layer-1 ratio (no SI unit needed)